PART 1 STEP 1

Entity	Attributes	Primary Key
Department	dept_id, dept_name	dept_id
Student	student_id, matric_no, full_name, dept_id	student_id
Lecturer	lecturer_id, lecturer_name, dept_id	lecturer_id
Course	course_id, course_code, course_title	course_id
Enrolment	enrolment_id, student_id, course_id, session	enrolment_id

- **Departments and Students:** One student belongs to only one department, while one department can have many students.
- **Students and Courses:** One student can register for many courses,
- and one course can have many students. Because this creates a many-to-many ($M:N$) relationship, it cannot be stored directly and requires a linking (junction) table like *Enrolment*.
- **Courses and Lecturers:** One course can have multiple (many) lecturers across different sessions, and one lecturer can teach many courses

PART 2 STEP 1: First Normal Form (1NF)

Removed multi-valued cells so that every cell contains only a single atomic value, and established a composite primary key consisting of matric_no and course_code.

matric_no	student_name	student_dept	course_code	course_title	lecturer	lecturer_dept
20231234	Ada Obi	Software Engineering	SEN307	Database Systems	Dr A. Okafor	Software Eng.
20231234	Ada Obi	Software Engineering	SEN301	Software Architecture	Dr M. Bello	Software Eng.
20231235	John Musa	Computer Science	SEN307	Database Systems	Dr A. Okafor	Software Eng.
20231235	John Musa	Computer Science	COS301	Operating Systems	Mrs F. Eze	Compute r Sci.
20231236	Grace Okon	Software Engineering	SEN301	Software Architecture	Dr. M. Bello	Software Eng.

PART 2 STEP 2: Second Normal Form (2NF)

The Resulting 2NF Tables:

Removed partial dependencies by splitting attributes that depended on only part of the composite primary key into separate Student and Course tables, leaving the Enrolment table with fully dependent attributes

Student Table (matric_no is Primary Key)

matric_no	student_name	student_dept
20231234	Ada Obi	Software Engineering
20231235	John Musa	Computer Science
20231236	Grace Okon	Software Engineering

Course Table (course_code is Primary Key)

course_code	course_title
SEN307	Database Systems
SEN301	Software Architecture
COS301	Operating Systems

Enrolment Table (matric_no + course_code form the Composite Primary Key)

matric_no	course_code	lecturer	lecturer_dept	session
20231234	SEN307	Dr A. Okafor	Software Eng.	2026/2027
20231234	SEN301	Dr M. Bello	Software Eng.	2026/2027
20231235	SEN307	Dr A. Okafor	Software Eng.	2026/2027

matric_no	course_code	lecturer	lecturer_dept	session
20231235	COS301	Mrs F. Eze	Computer Sci.	2026/2027
20231236	SEN301	Dr M. Bello	Software Eng.	2026/2027

Step 3: Third Normal Form (3NF)

Removed transitive dependencies by extracting department and lecturer details into separate lookup tables and linking them using foreign keys

Step 4: Final Schema Format

DEPARTMENT (`dept_id` PK, `dept_name`)

STUDENT (`student_id` PK, `matric_no`, `full_name`, `dept_id` FK Department)

COURSE (`course_id` PK, `course_code`, `course_title`)

LECTURER (`lecturer_id` PK, `lecturer_name`, `dept_id` FK Department)

ENROLMENT (`enrolment_id` PK, `student_id` FK Student, `course_id` FK Course, `lecturer_id` FK Lecturer, `session`)